

Question 8 – Measuring Luminosity Using an Oil Spot Photometer

(a) Show that $\frac{L_2}{d_2^2}$ is constant using all data

✔ 3 marks

We are told to verify that the quantity $\frac{L_2}{d_2^2}$ is constant.

L_2 (units)	d_2 (m)	L_2/d_2^2 (units/m ²)
5.00	0.350	40.8
10.00	0.495	40.8
15.00	0.606	40.8
20.00	0.700	40.8

Since the value is the same for each pair:

$\frac{L_2}{d_2^2} = 40.8 \text{ units/m}^2 \Rightarrow \text{A constant}$

✔ Conclusion: The relationship is confirmed — brightness is inversely proportional to distance squared.

(b) Why use small spherical lamps?

✔ 1 mark

✔ Answer:

Small spherical lamps can be approximated as **point sources**, ensuring light radiates evenly in all directions for reliable comparison.

(c)(i) Determine mean value of d

✔ 1 mark

Measured values (in metres):
0.88, 0.86, 0.90, 0.89, 0.86

$$\bar{d} = \frac{0.88 + 0.86 + 0.90 + 0.89 + 0.86}{5} = 0.88 \text{ m}$$

(c)(ii) Calculate random uncertainty in the mean value of d

✔ 2 marks

Use range/2:

$$\text{Range} = 0.90 - 0.86 = 0.04 \Rightarrow \text{Uncertainty} = \frac{0.04}{2} = 0.02 \text{ m}$$

🧠 Marking guidance also accepts:

$$\text{Uncertainty} = \frac{\text{max} - \text{min}}{n} = \frac{0.04}{5} = 0.01 \text{ m}$$

Either is valid depending on method used.

(c)(iii) Determine the luminosity of the Sun

✔ 2 marks

Use:

$$\frac{L_{\text{lamp}}}{d^2} = \frac{L_{\text{Sun}}}{D^2} \Rightarrow L_{\text{Sun}} = L_{\text{lamp}} \cdot \frac{D^2}{d^2}$$

Given:

- $L_{\text{lamp}} = 1.0 \times 10^2 \text{ W}$
- $d = 0.88 \text{ m}$
- $D = 1.5 \times 10^{11} \text{ m}$

$$L_{\text{Sun}} = 100 \cdot \left(\frac{1.5 \times 10^{11}}{0.88} \right)^2 = 2.9 \times 10^{26} \text{ W}$$

🧠 Revision Tips

- Brightness of a source follows an **inverse square law**:

$$\text{Brightness} \propto \frac{1}{\text{distance}^2}$$

- The oil spot photometer relies on matching brightness from two sources.
- Always quote values to **appropriate sig. figs**, especially with large powers of 10.
- Point sources** simplify the geometry of light propagation.